How to Design an Effective AI Governance Operating Model

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Initiatives: Establish Robust AI Governance and Organizational Trust; Strategy, Risks and Opportunities; Technologies and Markets

Data and analytics leaders can use this research to design an AI governance operating model for their organization's business objectives. This model should connect with other governance bodies and extend existing governance models to AI-specific considerations of trust, transparency and diversity.

More on This Topic

This is part of an in-depth collection of research. See the collection:

- Scenario Guide to AI Essentials for Digital Leaders

Overview

Key Findings

- AI governance provides risk mitigation, faster time to value and better utilization of resources. AI governance is non-negotiable because AI exists throughout the enterprise, necessitating balanced AI value and risks.
- The three common characteristics of AI governance — trust, transparency and diversity — often individually appear in AI governance strategies but seldom appear all together. This impairs AI's effectiveness and safety.
- Generative AI (GenAI) is increasingly in need of AI governance, placing it among the top three positive impacts on broader AI brought by GenAI.

Recommendations

Data and analytics (D&A) leaders should:

- Adopt Gartner's D&A governance operating model as a starting point for the AI governance model. To accommodate AI, repurpose core governance capabilities or develop new ones.

- Extend basic governance capabilities by adding AI-specific pillars: transparency, diversity and trust.

- Implement your target AI governance operating model by mapping governance pillars to key AI components based on business priorities and expected outcomes for AI use cases. The differentiating AI capabilities must include an ethical foundation and AI life cycle management.

Strategic Planning Assumptions

- By 2027, AI governance will become a requirement of all sovereign AI laws and regulations worldwide.

- By 2027, AI governance and responsible AI capabilities will be part of 75% of AI platforms, making them the main area of AI competition.

Introduction

Now that the AI surface area is dramatically expanding in enterprises and a responsible use of AI is a societal and CEO concern, D&A leaders are asking about AI governance, even before AI implementation. They want to know how to weigh AI's potential value against the compliance and reputation risks that such technology introduces.

Definition: AI governance is the process of creating policies, assigning decision rights and ensuring organizational accountability for the risks and decisions for the application and use of artificial intelligence techniques.

Governance of AI-enabled systems and processes should differ from traditional systems and processes but not be a brand-new, stand-alone undertaking. It should rely on the achieved successes in governing D&A or other areas with successful and recognizable governance approaches and operating models. D&A leaders should use the D&A governance framework as a starting point and add AI-specific pillars, see Figure 1.

Figure 1: Extending D&A Governance to AI-Specific Pillars

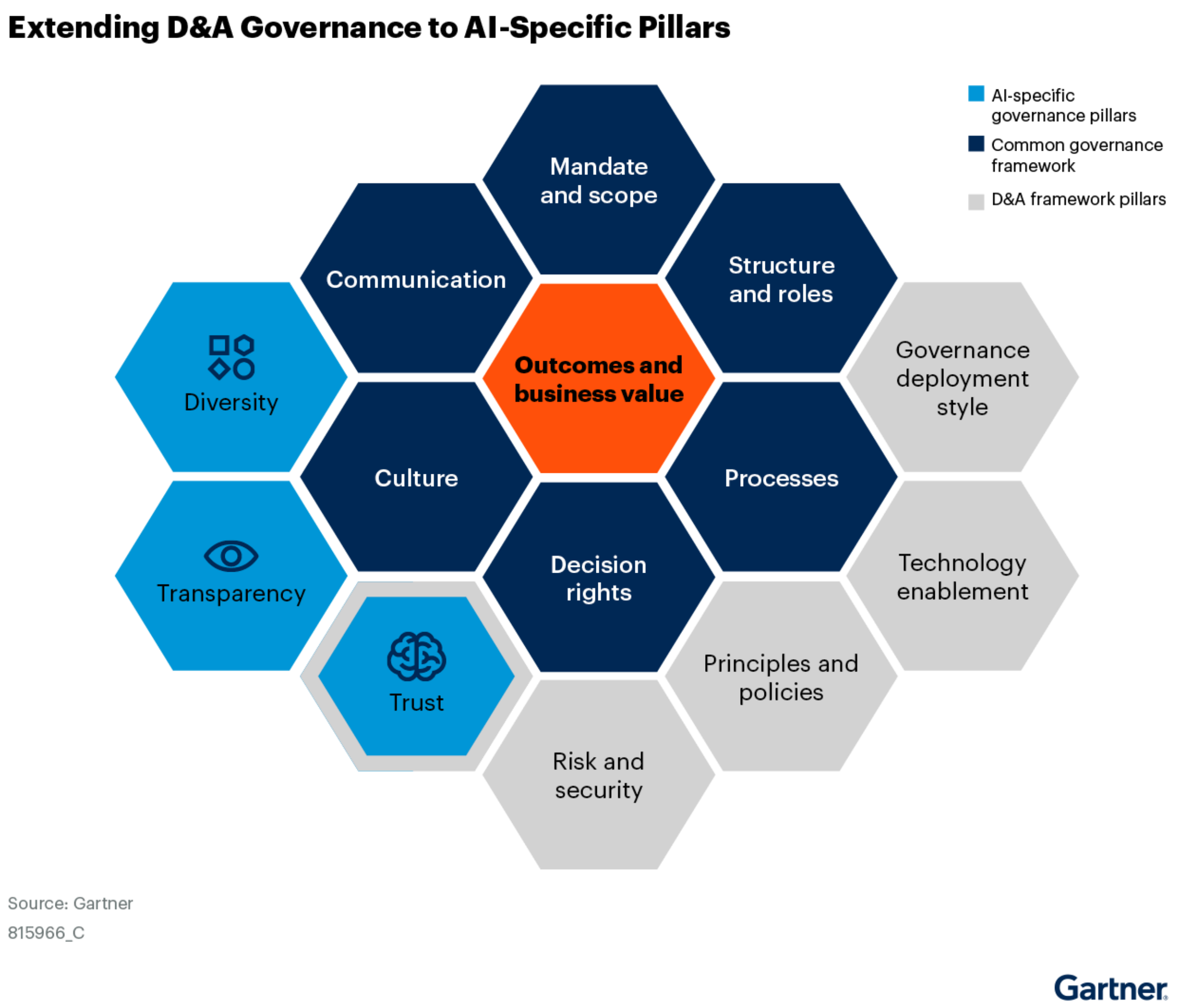


Figure 1 shows the full AI governance framework that includes:

- A **common governance framework** (typical for any governance) — It comprises six pillars: mandate and scope, structure and roles, processes, decision rights, culture, and communication. Artificial Intelligence Requires an Extended Governance Framework outlines how to map AI activities to this framework.
- The **D&A framework** pillars of trust, principles and policies, technology enablement, governance style, and risk and security — They are described in How to Design an Effective Data and Analytics Governance Operating Model.
- The specific pillars necessary for **AI governance**: trust, transparency and diversity — Trust in AI overlaps with D&A governance but goes further to trust in people's actions and AI techniques, as trustworthy AI positively affects AI adoption.

This document focuses on the governance framework's AI-specific pillars and the associated AI governance operating model.

Analysis

Extend Your D&A Governance Framework to AI-Specific Characteristics

Although AI governance shares some capabilities with other business function governance bodies, it contains unique characteristics: trust, transparency and diversity. Each of these characteristics apply to people, data and techniques. Think of it as an essential AI governance matrix as depicted in Table 1. It shows how the D&A governance framework could be extended to include the AI-specific components needed to complete the picture. (The best practice is to get D&A governance in place first. See Data and Analytics Essentials: D&A Governance.)

Table 1: AI Governance Matrix

↓	People ↓	Data ↓	Techniques ↓
Trust	AI adoption depends on the right level of trust in AI solutions.	Not all data is created equal. Some data sources are more trusted and authenticated than others, and context affects the meaning of data.	Explainability, bias mitigation and feedback loops improve trust in AI outcomes.
Transparency	The organization keeps track of and documents how AI decisions are made.	Metadata, provenance, observability, transparency of pipelines and visualization give an understanding of how data affects outcomes.	AI models vary from “black-box” to “glass-box;” select the right ones to meet regulatory, business and ethical requirements.
Diversity	Various perspectives are necessary to make decisions that impact individuals, groups, organizations and society.	Data must fully represent the problem being solved, including adversarial data for testing AI solutions.	Choose an appropriate technology, approach or provider for a use case. Use composite AI to broaden AI abstraction mechanisms.

Source: Gartner

Raise awareness of the questions to ask about data, techniques and people in AI projects, as described below in this section. Examining each element of the AI governance matrix is the core driver of AI governance activities. In AI, new questions arise regularly. For example, new concerns about AI-originated disinformation add questions — such as "how do we discern real content from deepfakes?" — that may require a new policy, guideline or standard. You can ask your own questions and use sample questions provided in this research or in other guidelines.

Examples include:

- Gartner's Tool: Generative AI Policy Kit
- U.S. GAO Artificial Intelligence: An Accountability Framework for Federal Agencies and Other Entities
- EU Artificial Intelligence — Questions and Answers

What should you do if your D&A governance program or operating model is missing or ineffective? Given the importance of the common foundation between data governance and AI governance, we recommend you adopt a modern, outcome-based D&A governance operating model as fast as possible (see 2024 Strategic Roadmap for Data and Analytics Governance).

Ensure Trust in AI Systems to Foster AI Adoption

Trust — both its presence and absence — in AI outcomes is the main barrier to AI adoption. People's attitudes to AI range from complete trust in inaccurate solutions to complete mistrust in accurate solutions. To shift people's views to a reasonable position within this spectrum, governance should raise awareness of AI's probabilistic nature. In other words, people need to understand that some AI outputs will be incorrect but most will be useful. As a best practice, provide an easy way for the users to report exceptions and their experiences with AI systems.

For sound AI governance, account for:

People's Trust

Humans decide what problems to solve, how to solve them and how to deliver the solutions. They should know what to expect.

Sample questions to consider:

- Did we clearly articulate the AI system's purpose and intent?
- How do we demonstrate the trustworthiness of outcomes?
- Do we set the right stakeholders' expectations that outputs improve over time?

Trust in Data

When extending D&A governance to AI, trust mechanisms developed for data could be reused for AI. Define distinct, graduated trust levels for data to distinguish fully trusted data from least trusted data. For example, you might assign the lowest trust level to data you know to be biased and some third-party data.

Raise awareness of the need to proceed with caution (and how to do so) but don't ban low-trust data outright. Apply governance precepts developed for D&A governance to verify, validate and fine-tune data. Validate new data with known models to check whether you get similar results. Demand rigor when separating training and validation data (see Leverage a Trust Model to Reset Data and Analytics Governance).

Sample questions to consider:

- What measures must we take to ensure data privacy and compliance with relevant regulations?
- If we share our data, how do we prove and maintain its trustworthiness?
- What level of data quality is necessary for this AI solution?

Trust in AI Techniques

Formulate guidelines for acquiring, training and modifying models. Sources of techniques and trained models could be your AI team, external consultants, commercial platforms, open-source libraries or, increasingly, services and APIs that deliver models. As with data sources, trust in the techniques varies.

Sample questions to consider:

- How are potential biases or discriminatory outcomes identified and addressed by the AI techniques employed for the use case?
- How do we validate an AI solution?

- How is the AI techniques' performance and effectiveness monitored and evaluated?

Achieve the Right Degree of Transparency to Reduce Risks and Build Confidence in AI

Lack of clarity about AI outcomes poses compliance, business and reputational risks. Transparency involves deliberate decisions, starting with the question: "Should we implement this use case?" For instance, implementing facial recognition may have different answers based on risk and value considerations. [1] Providing reasonable explanations of AI conclusions boosts audience confidence. Evaluating AI capabilities' impact on compliance, intellectual property and ethics for each use case is crucial and requires transparency guidelines.

For sound AI governance, we recommend developing transparency precepts in light of the following points:

People Transparency

Document decisions about AI. Make information about AI development available to affected employees, activists and representation organizations, such as IT groups, unions and customer advocacy groups. Governance guidelines should require documentation and code transparency.

Sample questions to consider:

- Are the models owned by our organization understandable to everyone, not just their creators?
- Are we transparent about efforts to prevent ethics-related risks?
- Have we documented how we made trade-offs and the alternatives we debated?

Data Transparency

AI-enabled capabilities, such as conversational AI, raise new questions about data. For example, what happens if sensitive data is read aloud by a chatbot? AI governance should establish tiers of data accountability, depending on the sensitivity of AI-enabled actions.

Data undergoes complex transformations within the data pipelines that feed AI systems. This results in “data derivatives” and risks that the transformed data’s initial meaning is lost, which could result in it being misinterpreted. AI governance should therefore extend existing data observability and metadata guidelines to AI initiatives. To provide data transparency, one of the first tasks of AI governance should be to develop standards for collaboration, data preparation, metadata and visualization tools.

Sample questions to consider:

- Are we clear about the data’s meaning?
- If data is refined or aggregated, do we have access to the initial raw data?
- Are the relevant staff dealing with AI systems properly trained to detect and manage bias in data?
- Do we understand access rights to the data we use?

Transparency of AI Techniques

Black-box algorithms are complex and not possible to interpret. In contrast, transparent “glass-box” algorithms are simpler. Transparency can be achieved through giving preference to glass boxes or by employing approaches for documenting, testing and validating complex models. When accuracy and interpretability are crucial, combining glass-box and black-box algorithms can provide transparency.

Issue provisions for the algorithms’ accuracy, simplicity, scalability and interpretability, as well as for how easy they are to understand. Require rigorous documentation, testing templates and embedded self-assessments in the development process. Governance policies should facilitate version control and the ability to trace the AI solutions’ history.

Sample questions to consider:

- Does our solution conform to regulatory requirements and account for any other business and process constraints?
- Can we enable continuity if an AI team member leaves our organization?
- Are the relevant staff dealing with AI systems properly trained to interpret AI model output and decisions?

Pursue Diversity in People, Data and Techniques to Enable Ethical and Accurate AI Outcomes

Diversity counteracts bias. Bias can lurk anywhere in the AI process. Data might represent just part of the picture, not the entire picture. The team could apply routine approaches when a fresh method would lead to better results. Output could be "good enough," not the best possible. Understanding of the need for diversity is often incomplete.

Diversity is required not just in terms of people's minds, backgrounds and cultures but also in data selection and technique choices. Establish guidelines to support multidimensional perspectives relevant to the defined scope. Demand representative data to cut AI bias. Diversify techniques to solve complex problems.

For sound AI governance, consider:

Diversity in People

Bring in diverse participants — permanent members of the governance organization or participating via collaboration — who give their perspectives on AI governance decisions and challenge the expected outcomes. Individuals representing a relevant variety of skills, knowledge areas, experiences, ages, genders, cultures, races and ethnicities bring unique perspectives to ask the right questions about AI use cases and to validate the AI outcomes. In rare situations, diversity could be counterproductive due to sensitivity or scope of an AI use case.

Sample questions to consider:

- Have we considered regional and cultural differences in delivering the outcomes?
- Did we engage relevant domain experts to validate outputs during the AI system development?
- Are any groups at an advantage or disadvantage as a result of this output?
- Are there any situations where diversity is harmful?

Data Diversity

Data for AI often contains incomplete or biased information because data sources are insufficiently diverse. Diverse data enables better accuracy. Missing data impairs results. Recognize biases in training data, and improve training data with additional information derived from various data sources.

In some cases, it is not possible to collect data — for reasons of privacy, for example. In this situation, awareness of missing data and the various methods of mitigating such omissions is necessary. In other cases, data must be complete by law. You cannot, for example, withhold data for a legal case.

Organizations may face situations where they can start collecting missing data by implementing a minimum viable product (MVP) and improving it on the basis of data originating from that product use. Governance procedure should have provisions for releasing MVPs to differentiate with data.

Sample questions to consider:

- Are any groups at an advantage or disadvantage because of the data we use?
- Have we considered alternative data sources to have fuller information for this use case?
- When do we allow synthetic data to substitute real data?
- Have we found adversarial examples that could invalidate the AI model?
- Is data incomplete because of limited access?

Diversity of AI Techniques

AI techniques are reaching organizations in increasingly diverse forms — as machine learning libraries, APIs, enterprise applications and stand-alone applications with embedded algorithms, to name just a few. Data scientists are the best people to develop custom AI solutions. Many solutions could be bought, rather than developed from scratch. AI governance guidelines should not be limited to a single technology, approach or provider. Maximize your ability to reuse the models and elements of AI solutions. Define a governance process to evaluate business returns and help you decide either to iterate or move on to another AI project.

Sample questions to consider:

- Do we understand the inner workings of different techniques so we know which techniques to use to solve the problem with its constraints?
- Do we consider solutions in a one-sided way? Are we aware of the latest innovations?

- Have we considered alternative solutions for the problem?

Map AI Governance Pillars to Key Operating Model Components

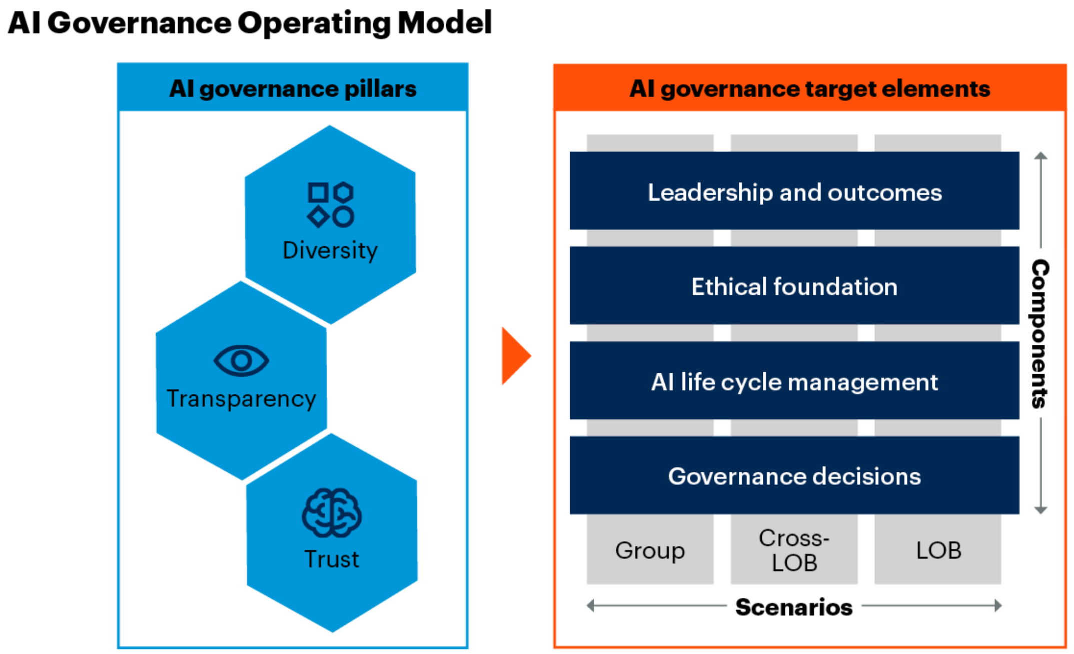
AI is probabilistic, which is hard for people used to deterministic thinking. This is the main driver of AI governance that should be concerned with anticipating and mitigating risks, manifested in the following:

- Unexpected outcomes might not be trustworthy; they often cannot be reproduced as the model outcomes are nondeterministic.
- Exceptions and their processing must be part of AI systems design, as the AI model accuracy is never 100%. Feedback loops must be required.
- Regulatory compliance with many new — effective and pending — laws for exploiting AI systems, such as the EU AI Act or the U.S. Executive Order on AI, is mandatory.

AI-enabled systems involve learning based on interactions and iterations, instead of requirements stated upfront. They require an operating model reflecting such considerations. Figure 2 maps the AI governance characteristics discussed earlier to four components in the AI governance target operating model:

- **Leadership and outcomes** — How should we organize, and what business outcomes must the organization achieve and why?
- **Ethical foundation** — Should we apply AI in a specific situation?
- **AI life cycle management** — What is the AI life cycle framework, and where in the life cycle are governance decisions made?
- **Governance decisions** — What governance decisions are needed to achieve these and by whom?

Figure 2: AI Governance Operating Model



Source: Gartner
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The model shown in Figure 2 illustrates scenarios where:

- A centralized group AI function provides strategic AI services.
- Cross-LOBs and central AI teams collaborate on specific AI initiatives.
- LOBs provide AI precepts specific to their operational areas.

According to the 2023 Gartner AI in the Enterprise Survey, AI capabilities are shifting from decentralization to consolidation within AI teams or a hybrid mode, when compared with results from the 2021 Gartner AI in Organizations Survey. [2] AI talent is mostly centralized (47%); centralization is significantly higher for North America (51%) compared to Europe (41%). A multitiered organizational model, where a centralized team (hub) collaborates with several decentralized teams (spokes) and communities is less prevalent (36%).

D&A leaders should use this AI operating model as the starting point for their organization and tailor it accordingly.

Leadership and Outcomes

AI leadership approaches and structures vary due to the novelty of AI governance. We observed the following organizational leadership structures for governance. You don't need all the structures. When creating governance structures, consider your AI scope, AI maturity, industry, regulatory compliance and culture. Some of these structures could be an extension of the existing D&A structures:

- **AI Governance Council** — A diverse group of stakeholders spearheads AI governance efforts; it should have direct working relationships with a D&A governance council. The AI governance council can be strategic, tactical and operational. It is often chaired by the CDAO, populated by representatives from the business domains and facilitated by the lead AI steward. For more information, see Quick Answer: Do Enterprises Need an AI Board?

- **Responsible AI Office** — This group focuses on protecting organizational and societal values, such as legal, human rights, fairness and more. It identifies current and emerging AI risks and opportunities that the organization should anticipate and prepare for.

- **AI Privacy and Compliance Office** — This office is similar to the Responsible AI Office but instead for privacy and compliance.

- **AI Center of Excellence (COE)** — The COE helps to centrally manage talent effectively. For AI, organizations must strike a balance between AI being developed and used in the context of business units, versus AI being governed and orchestrated from a more central level. A common approach is to set up an AI COE that works together with local initiatives, facilitating them with expertise, data, technology, operations and governance.

- **AI Lab** — A lab is similar to the AI COE but at a smaller scale. In some cases, an AI Lab is dedicated purely to innovation.

- **AI Risk Management** — This specialized vertical mitigates AI risks, or sometimes more narrowly, mitigates AI model risk. AI model validators often reside in a risk management organization.

- **AI Ethics Board** — A group of internal and, sometimes, external, members establish ethical principles and resolve ethical dilemmas.

- **AI Empathy Lab** — This lab has a capability to emulate an end user's or an affected audience's perception of AI for those creating the AI solutions. This gives the team firsthand experience with their deliverables.

Outcomes should be defined, and the governance framework's benefit should be clearly stated. Outcomes must be measurable. Good governance should aim for AI investment outcomes that support business strategy realization and benefit governance outcomes. Outcomes are achieved by applying the right guidance and principles. How these outcomes add value differ. Consider these examples:

- AI use-case prioritization guidelines could directly influence the business bottom line.
- Improved AI risk management may limit the downside of high-risk, low-reward decisions.
- Timely issued AI standards can save license and cross-LOB communication costs.
- AI life cycle process governance may rationalize the AI model portfolio by eliminating redundant models and replacing lower-performing models with higher-performing ones.

Design business-outcome-oriented AI solutions by assessing AI value and validating KPIs, as described in How to Calculate Business Value and Cost for Generative AI Use Cases. For example, if an AI solution is expected to scale to several LOBs, require a scalability evaluation upfront, as many pilots turn out to be costly at scale. This will affect the design of AI solutions. Look at the AI risk caused, for example by low-quality data and the need to upskill your employees to handle AI solutions and make AI-enabled decisions. Document the risk of inconsistent decisions to educate executives and stakeholders about decision risks.

Recommendations:

- Develop outcome-based AI use-case prioritization guidelines.
- Determine accountability for the outcomes and reflect it in the governance standards, principles and policies.
- Involve diverse participants to anticipate risks that could be economical, cultural, organizational or regulatory. Designate risk leadership to mitigate outcome-related risks.
- Issue and implement communication guidelines to support trust and confidence in AI outcomes. Provide performance and monitoring guidelines and policies to track AI value over time.

A feedback loop is required to learn about systems behavior and exceptions, so you can adjust expectations for AI value.

Ethical Foundation

Start with high-level principles and associated guidelines. Develop a solid ethical foundation over time, eventually operationalizing it. Although not all AI uses are ethically sensitive, many are. Examples include the behavior of autonomous cars, the impact of AI on jobs, the data collection needed to train AI-enabled systems and how people will be interacting with AI agents. The AI ethics discussion should be taking place upfront, considering that:

- AI-enabled systems are mostly rooted in the data to which they are being exposed during model development and fine-tuning. This is different from deterministic systems that are built to meet concrete requirements that drive a system's behavior.
- AI systems can propagate nondesirable behavior and bias, at scale. They can lead to unintended consequences. The logic of the model and the resulting output are not always predictable.

In developing AI governance, D&A leaders should:

- Reach agreement with stakeholders about relevant AI ethics guidelines. Start by looking at the five most common guidelines that others have used: being human-centric, being fair, offering explainability, being secure and being accountable. Keep monitoring whether the continued learning of the AI-enabled systems strays away from the guidelines.
- Organize training in ethics and run workshops to create awareness within all AI initiatives that AI design and implementation require an ethical mindset and clear accountability.
- Spend significant time within each project working out how to operationalize the guidelines, as this is not self-evident.

AI Life Cycle Management

To give all AI participants a clear understanding of each step, express the AI life cycle in the framework. Having such a framework will ensure that no substantial activity is missed in the life cycle. It will also enable mapping of AI governance standards, guidelines, policies and review gates on the repeatable basis.

For a sample AI life cycle framework, see Operationalize Machine Learning by Using Gartner's MLOps Framework. It supports the release, activation, monitoring, experimentation and performance tracking, management, reuse, updating, maintenance, version control, risk, and compliance management.

In developing AI governance, D&A leaders should:

- Support an uninterrupted flow between the development, operationalization and full maintenance of AI models. Ensure all the appropriate gates and controls are followed during the AI life cycle. When possible, facilitate automated reviews against the defined technical and risk KPIs, thresholds, and remediation actions.
- Implement a comprehensive model governance process. Facilitate a robust and secure AI life cycle through guidelines and standards for model inventory, AI life cycle automation, and comprehensive model monitoring.
- Create new roles or responsibilities to manage AI artifacts, reviews, testing and scoring metrics in the model governance process.
- At the early stages of AI adoption, tie policies to the AI platform in a prescribed way to guarantee the platform works. Gradually, allow more freedom by decoupling the platform and policies.

Governance Decisions

This component addresses the AI-related governance decisions needed to achieve business outcomes, based on agreed-upon policies, accountabilities and decision rights. The AI specifics involve decisions that are context-sensitive and require trade-offs. For example, opaque black-box models require trade-offs between the lack of explainability and other parameters important to the business, especially for sensitive decisions in housing, credit, employment, education and healthcare. The trade-offs could change for different use cases depending on the use-case context.

Take the following steps to design this component of your operating model:

- Document governance decisions for transparency. This is especially important in the absence of clear government regulations.
- Define levels of use-case criticality to concentrate decisions on the most critical AI content and govern it aggressively. Provide mechanisms for critical decisions on a use-case basis. Allow greater autonomy in decisions for noncritical AI content.
- For each critical use case, require a project manager or a product manager who communicates with the governance organization. This person should put together diverse experts and recipients of the use case who can ask value- and risk-related questions about this use case. The governance council, or a similar organization, should debate alternatives and make decisions by answering such questions.
- Determine accountability for the people and groups entitled to make business, technology and ethical decisions. Ensure you have guidelines for expected accountability of external AI providers; this will facilitate the AI purchasing decisions. Governance organizations should think critically about what meaningful accountability entails for different use cases. For example, if things go wrong, it is useful to have a predefined escalation procedure to identify whom to contact for issue mitigations.

Evidence

[1] For an example of in-depth deliberation on facial recognition, refer to Bringing Facial Recognition Systems To Light by the Partnership on AI.

[2] **2023 Gartner AI in the Enterprise Survey.** This study was conducted to understand the keys to successful AI implementations and their impact on the broader AI that has been brought by generative AI. The research was conducted online from 19 October through 21 December 2023 among 703 respondents from organizations in the U.S., Germany and the U.K. The main sample consisted of 645 out of the 703. Organizations were required to have developed or intended to deploy at least two AI initiatives within the next three years. Respondents were required to be part of the organization's corporate leadership or report to corporate leadership roles. Fifty-eight out of 703 are the business intelligence (BI) sample. Organizations were required to have developed or intended to deploy at least one AI initiative within the next three years. Respondents were required to be part of the organization's corporate leadership or report to corporate leadership roles or below (senior manager and above) and to be primarily responsible for BI in their organizations. Both the main sample and the BI sample respondents were required to have a high level of involvement with at least one AI initiative, and they were required to have one of the following roles when related to AI in their organizations: determine AI business objectives, measure the value derived from AI initiatives, or manage AI initiatives' development and implementation. Quotas among the main sample were established for company size and for industries to ensure a good representation across the sample. No quotas were established for the BI sample. Disclaimer: The results of this survey do not represent global findings or the market as a whole, but reflect the sentiments of the respondents and companies surveyed.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

Toolkit: Setting Up a Committee Charter for AI Governance

A Checklist for Managing Risk Across Six Pillars of AI Governance

A Journey Guide to Manage AI Governance, Trust, Risk and Security

Reference Guide for AI Governance

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